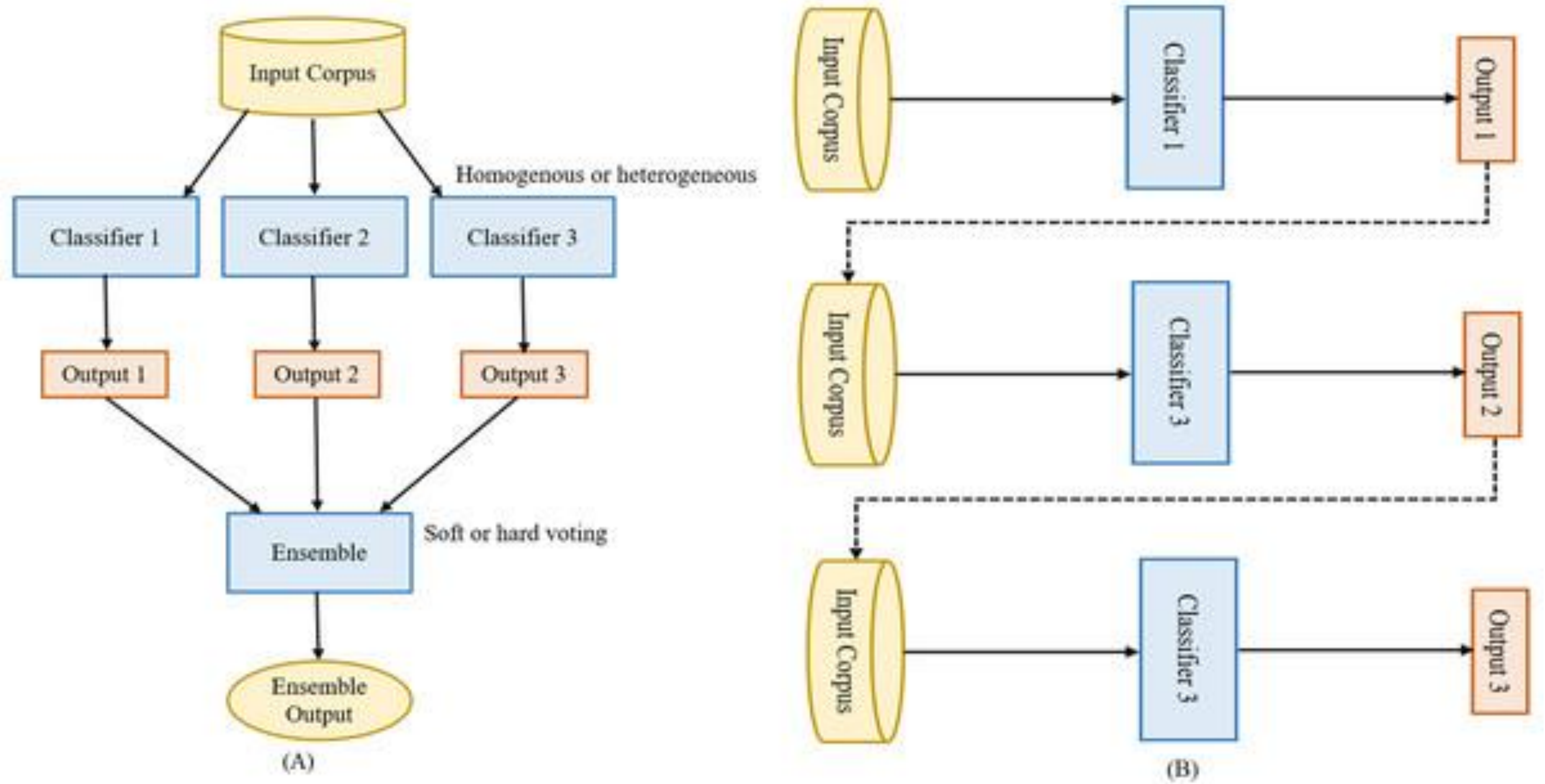
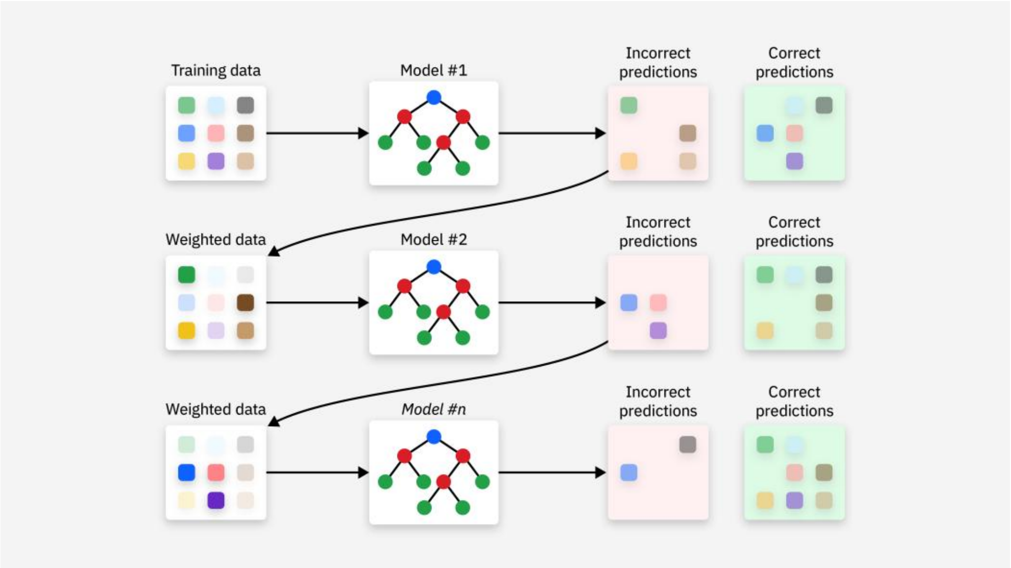


Ensemble Learning (The Power of Crowds)

WHY MANY MODELS ARE BETTER THAN ONE

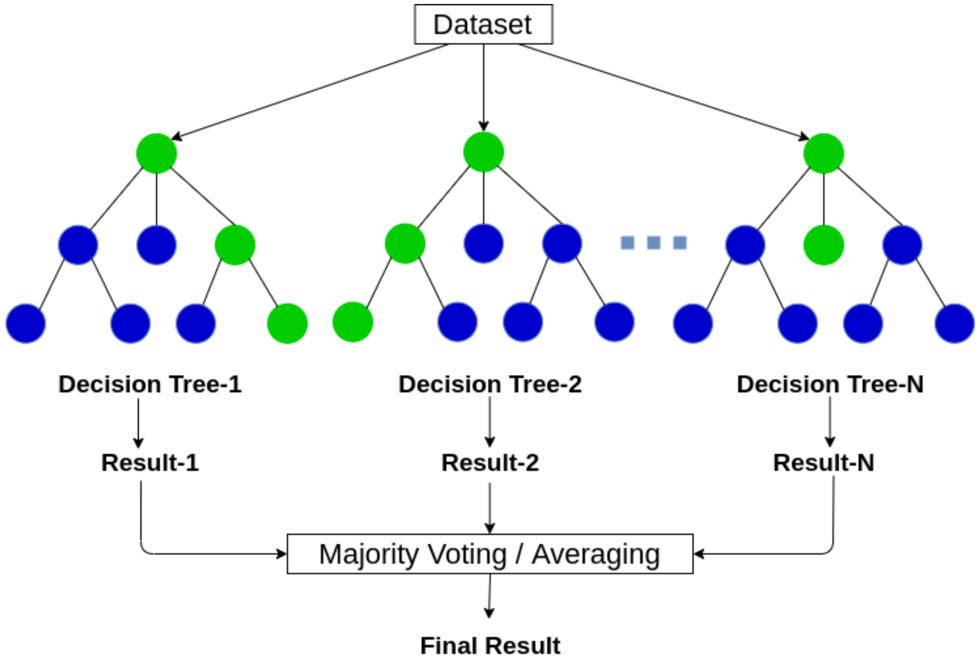
INSTRUCTOR: ZARYAB AHMAD KHAN

What is Ensemble Learning?



- **The Concept**: It is the idea that 100 average models voting together are significantly more accurate than one highly intelligent model alone.
- **Analogy**: A "Jury" in court. One judge might have a bias, but 12 jurors voting together are more likely to reach a fair verdict.
- **The Goal**: Reduce errors and prevent overfitting by averaging out the mistakes of individual models.

Random Forest (The Forest for the Trees)



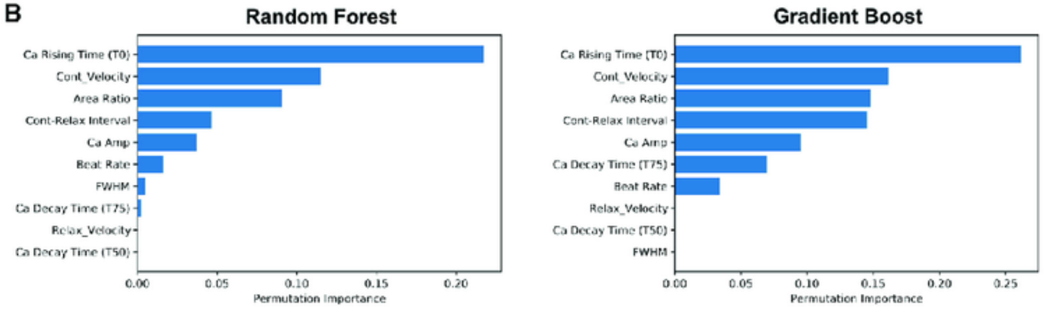
- **Definition**: A Random Forest works by building hundreds of different Decision Trees at the same time.
- **The Process**: Each tree is trained on a slightly different random slice of the data.
- **The Prediction**: For classification, the trees "vote" and the majority wins; for regression, the model takes the average of all tree predictions.

Feature Importance

A

Random Forest	Accuracy	F1 Score	Precision	Recall
Geometry	0.79	0.79	0.83	0.78
tSNE	0.88	0.87	0.89	0.88
UMAP	0.88	0.87	0.89	0.88
TriMAP	0.86	0.85	0.84	0.88
PacMAP	0.88	0.89	0.89	0.91

Gradient Boost	Accuracy	F1 Score	Precision	Recall
Geometry	0.76	0.72	0.88	0.89
tSNE	0.95	0.95	0.95	0.95
UMAP	0.95	0.95	0.95	0.95
TriMAP	0.86	0.87	0.9	0.88
PacMAP	0.86	0.87	0.88	0.89



- **The Question**: "Which columns of my data actually matter?"
- **The Tool**: Random Forests can mathematically rank which columns of data were the most helpful in making the final decision.
- **Usage**: This helps us understand the "why" behind a prediction and allows us to remove useless data columns in the future.